Monopolar EOG Data Recorded in Isotropic Directions

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1 Experimental Paradigm

This dataset comprises electrooculography (EOG) data recorded from fourteen healthy participants (5 males and 9 females; mean age 26.6±5.8 years), having normal or corrected-to-normal vision. The adopted eye movement data acquisition protocol was approved by the University Research Ethics Committee (UREC) at the University of Malta and each subject provided their informed consent before each recording session. Each subject was seated approximately 60 cm away from a 24-inch LCD monitor and was instructed to rest their head on a chin-and-forehead rest to ensure that stationary head conditions were maintained during data acquisition.

The subjects were asked to fixate their point of gaze (POG) on a circular cue that was shown on prescribed positions on the screen. The cue was initially shown at the centre of the screen, after which a sequence of 4-second trials commenced. In the first 1-second interval of each trial, the cue shifted to a specific position as discussed in the next paragraph, prompting the subject to make a saccade from the centre of the screen towards the indicated target. In the following 1-second interval, the cue was shown at the centre, prompting the subject to execute a return saccade back to the centre of the screen. Subjects were instructed not to blink during the first and second 1-second intervals. Finally, in the last 2-second interval, the cue remained at the centre but changed colour, prompting the participant to blink once.

The cue positions during the first 1-second interval of each trial were selected such that (i) the user carried out saccades in directions which were equiangularly spaced at 10° intervals around the centre of the screen, resulting in a total of 36 distinct directions, and (ii) all gaze displacement had the same magnitude of 12°. To prevent participants from anticipating movements, the cues were presented in a random sequence, rather than sequentially. In total, 288 trials were recorded from each subject, divided in four identical exercises with rest breaks in between.

Monopolar EOG signals were recorded using an 18-electrode setup, as shown in Figure 1. The setup included a ground and a reference electrode, placed on the forehead and on the left mastoid, respectively. To achieve consistent electrode placement across subjects, the other 16 electrodes were equiangularly spaced around the eyes at 30° angular intervals, as shown in Figure 1. Such electrode placement was facilitated using paper templates shaped like eye masks which included markings at the required 30° angular intervals originating from two reference points, which were spaced according to each subject's individual

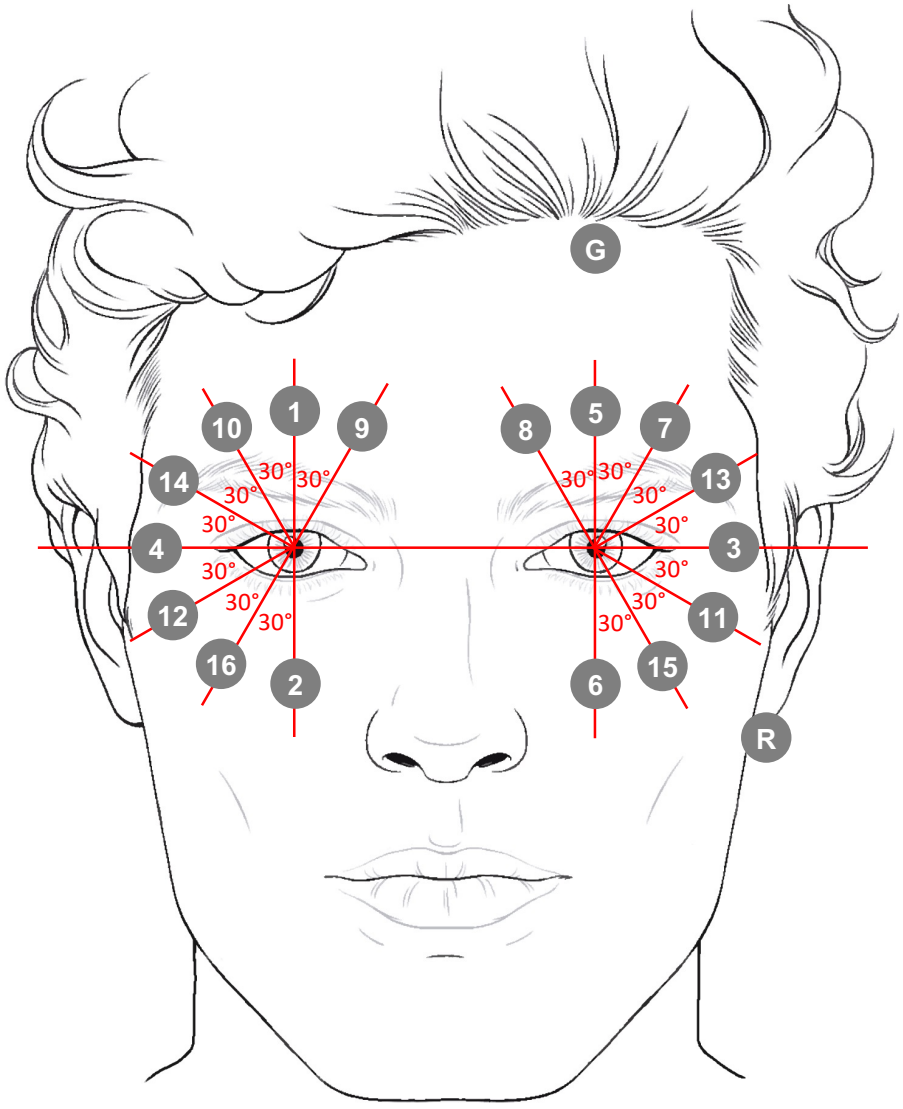


Figure 1: Electrode configuration used in this work. Image by Vectonauta on freepik.com, which was adapted for this work.

inter-pupillary distance. As a result, multiple eye masks were developed to accommodate varying inter-pupillary distances across subjects.

These electrodes were connected to a g.tec g.USBamp bio-signal amplifier, which recorded EOG signals at a sampling frequency of 256 Hz. A 30 Hz low-pass filter and a 50 Hz notch filter were applied.

2 Data Format

The data is provided in `MATLAB` (`*.mat`) format. The data recorded for each subject X is provided in separate folders, which are named `SX`. In each folder, the following files are provided:

- `EOG.mat`: This stores the recorded monopolar EOG signals. Specifically, the first row comprises the EOG signal recorded by electrode 1, the second row comprises the EOG signal recorded by electrode 2, etc.
- `ControlSignal.mat`: This file stores a control signal, of the same size as `EOG.mat`, where each sample contains a value, '1', '2' or '3', which identifies whether that particular sample corresponds to the first one-second interval of the trial ('1'), the second one-second interval of the trial ('2'), or the last two-second interval of the trial ('3').
- `Target_GA_stream.mat`: This file contains the gaze angles corresponding to the target cues on the screen that were shown to the user at each time sample. Specifically, the horizontal gaze angles are in the first row and the vertical gaze angles are in the second row, both in degrees.
- `PD.mat`: This file contains the inter-pupillary distance of the subject, in metres.
- `Dist_screen.mat`: This file contains the distance of the subject from the screen, in metres.

3 Reference

N. Barbara, T. A. Camilleri, and K. P. Camilleri, "A Systematic Quantitative Analysis on Bipolar Channel Selection for EOG-Based Gaze Displacement Estimation," *Biomedical Signal Processing and Control*.